



COURSE DESCRIPTION FORM

INSTITUTION National University of Computer & Emerging Sciences (FAST-NUCES) Karachi

PROGRAM (S) TO BE

Course Outline – SPRING 2026

EVALUATED

A. Course Description

Course	AI4015
Cours	Agentic AI
Credit Hours	3
Prerequisites Course(s) and by Topics	Programming (Preferably Python), Artificial intelligence, Machine Learning, Deep Generative AI, Learning, Natural Language Processing
Assessment Instruments Weights (homework, quizzes, midterms, final, programming assignments, labetc.)	Assignments 5% Mid-1 Exam 15% Mid-2 Exam 15% Quiz 5% work, Project 10% End-term Exam (Final) 50%
Course Coord	Dr. Muhammad Farrukh Shahid
URL (if any)/	mfarrukh.shahid@nu.edu.pk
Current Course Descriptio n	Agentic AI
Textbook(or Laboratory Manualfor Laboratory Courses)	1. Building Agentic AI Systems Anjanava Biswas Wrick Talukdar 2. Agentic Artificial Intelligence Pascal Bornet, Jochen
Reference	3. D. B. Acharya, K. Kuppan and B. Divya, "Agentic AI: Autonomous Intelligence for Complex Goals—A Comprehensive Survey," in IEEE Access, vol.13, pp. 18912-18936, 2025, doi: 10.1109/ACCESS.2025.3532853. 4. AI Agents vs. Agentic AI: A Conceptual Taxonomy, Applications and Challenges, https://arxiv.org/abs/2505.10468

Course	1. Provide the student with knowledge of the theories and applications of Agentic AI, brief description about Generative AI, AI agents and applications of agentic AI. 2. Familiarize the student with important concepts of self-governance (Self-organization, Self-adaptation), agency, and autonomy of agentic AI systems, ethics in agentic AI. 3. Familiarize student with the technical framework of agentic AI developments such as CrewAI, AutoGPT Taskade, n8n, Langflow y LangGraph.
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	A. Course Learning Outcomes (CLOs)

Course	1. <i>Describe different techniques in making Agentic AI in various scenarios</i> [Bloom's Taxonomy Level: 3, Learning Domain: Cognitive] 2. <i>Solve mathematical optimization problems pertaining to Agentic AI</i> [Bloom's Taxonomy Level: 3, Learning Domain: Cognitive] 3. <i>Discuss how a Agentic AI system should be evaluated in terms of the system's performance and the user's satisfaction with the system</i> [Bloom's Taxonomy Level: 6, Learning Domain: Cognitive] 4. <i>Learn about advanced topics and current applications of Agentic AI systems in realms like social networks and communities</i> [Bloom's Taxonomy Level: 6, Learning Domain: Cognitive] B. Program Learning Outcomes 1. Computing Knowledge 2. Problem Analysis 3. Design/Develop Solutions 4. Investigation & Experimentation CLO-1 Apply knowledge of mathematics, natural sciences, computing fundamentals, and a computing specialization to the solution of complex computing problems. CLO-2 Identify, formulate, research literature, and analyze complex computing problems, reaching substantiated conclusions using first principles of mathematics, natural sciences, and computing sciences. CLO-3 Design solutions for complex computing problems and design systems, components, and processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. CLO-4 Conduct investigation of complex computing problems using research-based knowledge and research-based methods
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Topics Coveredthe Co withNumber of Lectur Each Topic(assume 15-weekinstruction and lecturesweek)	in Refer Appendix A three per																																																																																																								
Projects/Experiments the	Students will be given assignments related to the theory concepts theylearninthe classroom. Or some advanced topics which students are supposedtoexplorebythemselves. Course A project discussing issues related to the state-of-the-artwork related to AgenticAIwillbeassigned. Note that scope of the project should lie within the Agentic AI / AI agent domain.																																																																																																								
Programming Assignments the Course	Done in A few programming labs are given to apply the key concepts of Agentic AI.																																																																																																								
Class Time Spent on(hours)	Theory				Problem				Analysis SolutionDesign																																																																																																
	25%				35%				40%																																																																																																
Oral and WrittenCommunicatio	Every student is required <u>to submit at least 01</u> written <u>report of typically 6</u> pages (IEEEFormat)andto <u>make 01</u> oral presentations <u>of typically 15</u> minute’s duration. Include only material that isgraded for grammar, spelling, style, and so forth, as well as for technical content, completeness,andaccuracy.																																																																																																								
Late SubmissionPolic	Deadlines are meant to be strictly followed. Any late submission (without and validreasonandjustification/ evidence) will be penalized. The penalty will be 50%. Any delay of morethanaweek would mean ZERO credit in that assessment (assignments, project).																																																																																																								



Appendix A

Week	Topics	CLOS
Week	Introduction to the course Introduction to Agentic AI The evolution of conversational AI towards autonomous agents, the 1 agentic AI and agentic workflows, from Is that AI? to When is there an AI agent?A From Automation to Autonomy, Comparison of classical agents,landscape transitio Applications of agentic AI Discussion on state-of-the-art Agentic AI projects	to1 AI
Week	2 Core Characteristics of AI Agents, Foundational Models: Generative AI asPrecurs Gen AI / Traditional AI, Agentic AI Components, TechnicalFoundation of Agentic ARCHITECTURAL APPROACHES in Agentic	a 1 A
Week	3 The journey of LLMS, Encoder-decoder architecture, Attention Mechanism, transf learning, transformer architecture,	
Week	4 Auto-regressive models (GPT-2, and GPT-3), Types of Auto-regressive LanguageModels Assignment no 01 announcement and Quiz no 01 (3rdlecture of a week) Assignment submission with in 15 days after the announcement	2
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Week	5 Practical demonstration / Research Papers Exploration / Industrial Expert lecture Course Project discussion - Research directions	
Week	6 Mid - 01 Exam	
Week	7 Large Language Models: A Survey, GPT Family, Palm family,	
Week	8 HOW LLMS ARE BUILT ? Data filtering, data-cleaning, deduplication, tokenization. Fine tuning, alignment	

Week	9 Cost-Effective Training/Inference/Adaptation/Compression Low-Rank Adaption (LoRA), Knowledge Distillation, Performance Measures-Statistical Metrics(ROUGE, BLUE), Model-BasedMetrics Assignment no 02 announcement and Quiz no 02 (3 rd lecture of a week) Assignment submission with in 15 days after the announcement	3
Week	10 RAG, Known challenges of LLMs, Retrieval-Augmented Generation working benefits of Retrieval-Augmented Generation	
Week	11 Mid - 02 Exam	
Week	Model Context Protocol (MCP), MCP ARCHITECTURE (host, client, server) 12	3
Week	13 A2A, Core Functions of the A2A Protocol, Technical Advantages of the A2AProt	4
Week	14 Technical frameworks for agent development	

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	Langchain,LangGraph, CrewAI, n8n	
Week	15 Understanding the concept of tool use in agents, LLM agent's capability of leveraging resources, Tool and function calling, Defining tools for agent, Direct LLM integration tools Assignment no 03 announcement and Quiz no 03 (3rd lecture of a week) Assignment submission within 15 days after the announcement	4
Week	16 Trust, Safety, Ethics, and Applications Importance of trust in AI, Techniques for establishing trust, Dealing with uncertainties and biases.	4
Week	17 Final Exams	

Instructor Name Dr. Muhammad Farrukh Shahid

Instructor Signature _____

Date: January 28, 2026